

OpenSEP: An Open Source Subjective Experiment Platform

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Abstract

Subjective experiments driven by human visual perception for images aid in the development of related technologies such as analysis, compression, and transmission, which garners substantial attention and research interest. However, the general subjective experiment platform is relatively lacking, and verifying the reliability of annotation data is often difficult. In response to these challenges, an open source subjective experiment platform, namely OpenSEP, is proposed in this paper. Specifically, OpenSEP mainly includes a contrast mode sub-platform that displays dual stimuli, allowing for the simultaneous display of both source stimulus and distorted stimulus for subjective testing. Moreover, a scoring mode sub-platform that displays single stimulus is also provided in OpenSEP. In this mode, subjects can only score the distorted stimulus individually after sequentially viewing the source stimulus and all the distorted stimuli. Besides, OpenSEP constructs a cross-validation sub-platform integrated mainstream Just Noticeable Distortion (JND) algorithms. Within this sub-platform, the reliability of subjective annotation data can be verified based on existing JND algorithms, and the effectiveness of newly proposed modeling algorithms can also be validated. The open source library for OpenSEP is available at <https://openi.pcl.ac.cn/OpenDatasets/OpenSEP>

CCS Concepts

• **Software and its engineering** → **Software libraries and repositories.**

Keywords

Subjective Experiment, Just Noticeable Distortion, Image, Intra Coding, Open Source Software

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1 Introduction

As we know, images and videos are currently the most popular forms of media. However, the raw data of images and videos is enormous, especially in the recent prevalence of 3840×2160 (4K) and 7680×4320 (8K) formats. Consequently, techniques for simplifying or compressing raw data have become necessary, where efficient image and intra-frame video compression schemes typically involve some distortions [4, 6, 12, 15, 16, 18]. Moreover, distortions also frequently occur in other applications of image and video processing and transmission. Actually, in the human visual system, not all objective distortions are perceptible to human subjective perception [10]. Therefore, modeling human perception can provide effective guidance for improving the performance of related compression, processing, and transmission algorithms [9, 13, 17].

In recent years, plenty of works have focused on perceptually-driven image distortion modeling [8, 10, 11, 14]. As image display and presentation technologies continue to advance, e.g., the increasing popularity of 4K and 8K applications, corresponding perceptual modeling techniques must also innovate to adapt to these new demands. In essence, perceptual modeling involves the analysis and modeling of human-annotated results, which is inherently dependent on subjective experiment platforms. However, current subjective experiment platforms suffer from the following issues. Firstly, there is the lack of general subjective experiment platform for perception distortion annotation, which can flexibly adjust the annotation process according to different annotation requirements, e.g., the display order of distorted stimuli, the duration of the subjects' observation, and whether rollback is allowed. Additionally, beyond comparing differences between source stimulus and distorted stimuli, there is a significant demand for categorizing existing distorted stimuli into multiple levels. For instance, setting different distortion levels based on annotation results can guide the improvement and performance evaluation of image/video compression algorithms. Furthermore, with the advancement of machine learning, deep learning, and even large-scale models, there is a growing need for large-scale data annotation. The involvement of numerous subjects using different terminal devices for annotation makes it challenging to verify the reliability of the annotated data.

In response to the aforementioned challenges, this paper proposes a flexible and configurable open source subjective experiment

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Figure 1: The annotation interface of contrast mode sub-platform displayed dual-stimuli

platform, named OpenSEP, to facilitate the annotation of perceptual data. Specifically, the main contributions of this platform can be summered as follows: (1) OpenSEP primarily includes a contrast mode sub-platform that displays dual stimuli. Utilizing this sub-platform requires two monitors: the left monitor will consistently display the source stimulus, while the right monitor will display the distorted stimuli according to the preset order. After observation, subjects can choose whether to observe distortions between the source stimulus and the distorted stimuli. All of the subject's operations and information about the terminal monitors will be recorded in the log to facilitate subsequent verification and analysis. (2) OpenSEP also offers a scoring mode sub-platform that displays single stimulus. In the operation of this sub-platform, the source stimulus and all distorted stimuli are first played in a preset order and interval. After the playback is completed and the subjects have a comprehensive understanding of the distorted stimuli, they can then score all the distorted stimuli based on the preset score levels. (3) A cross-validation sub-platform is also implemented in OpenSEP, which integrates existing Just Noticeable Distortion (JND) algorithms. JND interval of the source stimulus is first calculated according to JND algorithm. Afterward, by comparing the JND interval with the specific distortion values of each distorted stimulus relative to the source stimulus, the reliability of subject annotation results can be effectively verified. Conversely, this sub-platform also can be used to validate the effectiveness of the new proposed perceptual distortion modeling algorithm.

2 Usage Procedures

OpenSEP mainly provides a contrast mode sub-platform displayed dual-stimulus to determine whether the distortion is observable, i.e., the annotation for Just Noticeable Distortion (JND). Recognizing the demand for categorizing distorted stimuli based on predefined distortion levels, particularly in applications that generate non-linear distortion, such as video compression, OpenSEP also offers a scoring mode sub-platform displayed single-stimulus. Moreover, a cross-validation sub-platform is designed to verify the reliability of annotation results or the effectiveness of related algorithms. In this section, usage procedures will be introduced sequentially.

2.1 Contrast Mode Sub-Platform

The usage procedures of the contrast mode sub-platform mainly involve the following four steps: data preparation, platform configuration, executing annotation, and data processing.

Data Preparation: The main task of the contrast mode sub-platform is to determine whether there are visible differences between the distorted stimuli and the source stimulus. Therefore,

during data preparation, each group of data requires a source stimulus, i.e., the original image, along with multiple distorted stimuli, i.e., corresponding distorted images. To control the display order of the distorted stimuli in annotation, the distorted images should be named in an orderly fashion according to their distorted levels, e.g., Image1_1, Image1_2, Image1_3, etc. Conversely, the original image should be named as Image1_0. The platform supports annotating multiple groups of images with different contents, but each group of images must be placed in a separate dedicated directory.

Platform Configuration: Before conducting data annotation, it is necessary to configure the annotation platform. All configuration information can be modified in the *TiffComputerApplication.exe.config*. Specifically, *pictureFolderPath* represents the root directory of different groups of stimuli. *pictureFolderOrder* specifies the annotation order of different groups of stimuli and should be configured based on the directory names of each group in the following format, e.g., Image1, Image2, Image3, etc. It should be noted that annotation for the next group of stimuli can only begin after the current group has been completed. *pictureOrder* defines the filenames of the source stimuli displayed on the left monitor for different groups, such as Image1_0, Image2_0, Image3_0, etc. *ForbiddenPrevious* controls whether rollback is allowed during annotation. If rollback is enabled, the subject can re-annotate previously viewed images. *setInterval* determines the observation time for the subject, measured in milliseconds. The annotation buttons, as shown in the bottom right corner of Figure 1, will only be displayed after the subject has completed the mandatory observation period. *logPath* specifies the output location for the annotation log. To accommodate different annotation needs, the display order of distorted stimuli within a group can be flexibly controlled in OpenSEP. Users can customize the order, such as sequential display or binary search display, by modifying the Python script located at the path specified by *pythonfilename*.

Executing Annotation: During the data annotation process, two identical monitors with the same display settings are required, and their resolution must be greater than or equal to that of the images being annotated. In the annotation process, the left monitor will fixedly display the source stimulus, while the right monitor will display the distorted stimuli. Once the subject annotates the current distorted stimulus, the right monitor will display the next distorted stimulus according to the display order defined in the Python script, until the JND stimulus for the current source stimulus is found. Upon completing the annotation for one group, the platform will automatically proceed to the next group, continuing this process until all annotation groups are completed.

Data Processing: The annotation log will record the resolutions of both monitors, as well as the relationship between the



Figure 2: The annotation interface of scoring mode sub-platform displayed single-stimulus

annotated distorted stimulus and the source stimulus (whether they are similar or different). Users can utilize the scripts provided in OpenSEP to process the annotation information in bulk. We offer the 3-sigma algorithm to filter out anomalous information and an algorithm based on Gaussian distribution to determine the JND stimulus position from multiple annotation results, facilitating the subject modeling algorithm.

2.2 Scoring Mode Sub-Platform

Utilizing the scoring mode sub-platform for subjective annotation also involves data preparation, platform configuration, executing annotation, and data processing. Considering that the data preparation step and data processing step are similar to the contrast mode sub-platform, this sub-section will focus primarily on the platform configuration and executing annotation stages.

Platform Configuration: The definitions of *pictureFolderPath*, *pictureFolderOrder*, *logPath*, *pictureOrder*, and *setInterval* are analogous to those in the contrast mode sub-platform. The python script referenced by *pythonfilename* also provides the presentation sequence for the next stimulus. The difference lies in *gradeLevel*, which is used to set the desired level of granularity for the distortion stimulus categorization.

Executing Annotation: Only one monitor is required during the data annotation. The resolution of this monitor must be equal to or greater than that of the image to be annotated, and it will also be recorded in the log to facilitate verification. During the annotation process of a group stimuli, the source stimulus and all distorted stimuli will be displayed sequentially, allowing the subject to gain a comprehensive understanding of all distorted situations. After the initial observation is completed, the platform will display all stimuli in sequence again. At this point, the subject can score the stimuli according to the configured rating scale, as shown in Figure 2. Only after completing one group of annotations can the next group begin. Similarly, in subsequent subjective experiments, the subject must view all stimuli before starting the scoring.

2.3 Cross-Validation Sub-Platform

OpenSEP also provides a cross-validation sub-platform to verify the reliability of annotation results. Specifically, during the utilization, the existing JND algorithm provided by OpenSEP should be executed firstly to obtain the JND Map of the source stimulus, i.e., JND intervals of the original image. Subsequently, each pixel in the distorted image will be checked by executing the Python script file *checkJNDInfo.py* based on the above JND map. Given that the human visual system cannot simultaneously perceive the global information of an image [5, 7], especially for images with ultra-high resolutions such as 1920×1080 (1080P), 4K, and 8K, OpenSEP

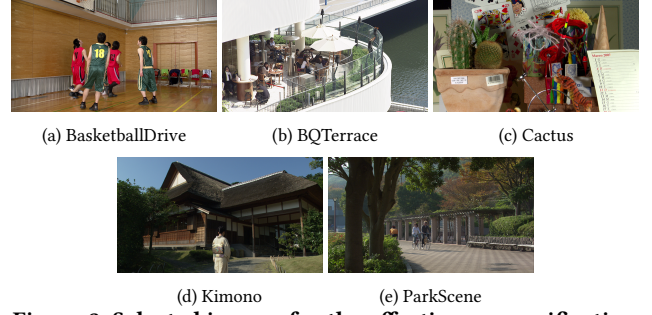


Figure 3: Selected images for the effectiveness verification employs a sliding window for pixel detection, with an adjustable window size. The proportion of pixels exceeding the JND interval within the window is recorded, and the maximum value among all windows is output as the distorted value of the current image relative to the original image, called perceptible distortion. A higher perceptible distortion indicates a greater likelihood of the corresponding distortion being perceived by human eyes. Ultimately, the perceptible distortion values of all images in the group of stimuli are output to a file, which can be used to validate the credibility of subjective annotation results. For example, if the subjectively annotated JND stimulus shows minimal perceptible distortion compared to the original image, it suggests that the corresponding annotation result is unreliable. Besides, based on the reliable annotation results and the opposite calculation process, cross-validation sub-platform can also be used to verify the effectiveness of the newly proposed JND modeling algorithms.

3 Experiments

3.1 Stimulus Generation

To validate the effectiveness of the proposed platform, we constructed a dataset and subjective experiments. As shown in Figure 3, five images with 1920×1080 resolution are selected from test sequences of JVET Common Test Condition (CTC) [3], and VTM-21.2 [1], the reference software of Versatile Video Coding (VVC) standard, is utilized to compression and generated distortion. Specifically, under the compression configuration of CTC, Quantization Parameters (QP) from 0 to 51 are used to perform intra compression of above five images, where the distortions increase with the QP value. The image compressed by QP 0 is employed as the source stimulus, while the corresponding image compressed by QP 1 to QP 51 are used as distorted stimuli. As a result, the subjective annotation task in each group of stimuli is to identify the image within the distorted stimuli set, where the distortion relative to the source stimulus becomes just noticeable.

3.2 Subjective Validation

Our subjective testing adheres to the recommendations of BT.500-13 [2]. The test terminal is computer running Windows10, with two monitors set to the resolution of 1920×1080 . The experiment platform is the proposed contrast mode platform in OpenSEP. The observation duration for subjects is set to 3000 milliseconds. We selected 30 subjects aged between 18 and 50, randomly dividing them into two equal cohorts for subjective experiments to perform cross-testing. Prior to annotation, we train the subjects to ensure they understand the experiment's purpose and the platform's operation.

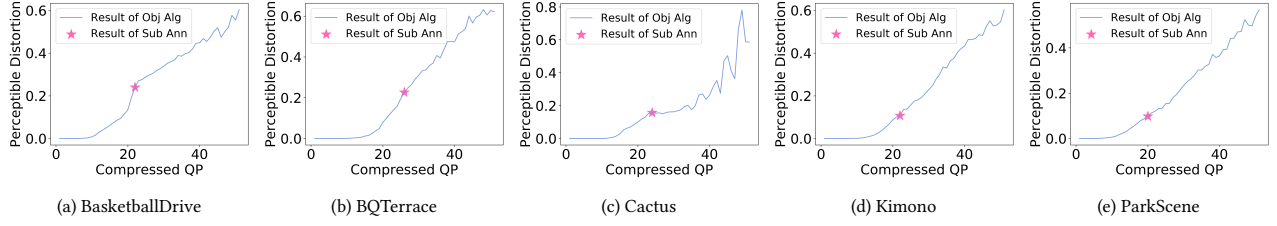


Figure 4: Perceptible distortions and JND positions of the five group stimuli generated by the selected images

Table 1: Correlations between the two cohorts of final annotation results. $\uparrow/\downarrow$ denote that the higher/lower value, the higher correlation

	SROCC $\uparrow$	PLCC $\uparrow$	KROCC $\uparrow$	RMSE $\downarrow$
Cohort1 vs. Cohort2	0.9250	0.9631	0.8819	0.0310

After the subjective experiment, 3-sigma algorithm in OpenSEP first filtered outliers from the annotation results of both cohorts. Note that if an outlier is detected, all annotations from that subject are removed. Subsequently, we calculate the mean position (QP from 1 to 51) of the Just Noticeable Difference (JND) stimulus for each source stimulus based on a Gaussian distribution to obtain the final JND position. We further calculate the correlation between the results of the two cohorts according to the widely used metric, including Spearman Rank-Order Correlation Coefficient (SROCC), Pearson Linear Correlation Coefficient (PLCC), Kendall Rank-Order Correlation Coefficient (KROCC), and Root Mean Square Error (RMSE), as shown in Table 1. The high correlation convincingly demonstrates the effectiveness of the proposed platform.

3.3 Cross-Validation

We also utilized the cross-validation sub-platform in OpenSEP to verify the subjective annotation results from the previous subsection. Specifically, the JND algorithm [14] provided in OpenSEP is used to generate the JND Map. During the calculation of perceptible distortion for each distorted stimulus, the sliding window size is set to 128×128 . All the annotation results from the 30 subjects are merged, and JND positions are recalculated. The final perceptible distortions calculated by the objective JND algorithm (Obj Alg) and the JND position counted from subjective annotations (Sub Ann) are shown in Figure 4. It can be observed that as compression distortion increases (due to the increase of QP values), perceptible distortion significantly increases. The annotated JND positions from the subjective experiment all appear in regions with substantial perceptual distortions. This indicates that the annotation results from the subjective experiment are relatively reliable.

4 Conclusion

Considering the lack of general subjective experiment platform and the difficulty in verifying the reliability of annotation results, this paper proposes an open-source subjective experiment platform named OpenSEP. OpenSEP primarily includes a contrast mode sub-platform that displays dual stimuli, allowing subjects to simultaneously observe both the source stimulus and the distorted stimulus and annotate whether they perceive any visual distortion. Additionally, a scoring mode sub-platform displayed single-stimulus is also provided. During the annotation process, subjects can score the

distorted stimulus only after observing all stimuli and understanding the overall distortion situations. Moreover, OpenSEP offers a cross-validation sub-platform, which can be utilized to assess the reliability of subjective annotation results and the effectiveness of the newly proposed JND modeling algorithms. To validate the effectiveness of OpenSEP, we conducted a subjective experiment involving 30 participants and verified the annotation results.

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